Outline for Lesson

- ★ **Monday, 4/29 10:20-11:20, Intro to Astronomy**
- ★ **Objective:**
  - To further the students' knowledge of exoplanets by introducing the concept of transits. Students will be encouraged to apply a basic understanding of exoplanets to the new information presented in this lesson. The main goals are for the students to think critically and recall the main ideas about exoplanets and transit detection.
  - Demonstrated through PowerPoint and interactive opportunities, such as questions and a worksheet
- ★ **Slide 1: Introduction**
  - Set/Hook -> Transition from Dr. Rodruck's lecture to mine
  - Call on ~3 people to recall one fact about exoplanets
    - Reward with moonpies!!!
- ★ **Slide 2: What are Exoplanets?**
  - 1000s+ exoplanets confirmed from a small sample of our galaxy
    - Confirmed vs. Candidate
    - Closest exoplanet is about 4 light-years away
  - Types:
    - Gas Giants
      - Gas with a solid core
      - Jupiter-sized or bigger
      - Often close to their host star
    - Neptunian/Neptune-like
      - Closer to Neptune/Uranus size
    - Super-Earth
      - Larger than Earth
      - Gas or rocky
    - Terrestrial
      - Rocky
      - Earth-sized or smaller
  - Most of the exoplanets that have been discovered are located in a small region in our galaxy
  - More planets than stars in the Milky Way! (NASA's Kepler Space telescope)
- ★ **Slide 3: Methods of Discovery**
  - Radial Velocity
    - Star "wobbles" from a gravitational tug-of-war between an orbiting planet and its star
    - Indicates if a star has planets and how big/many there are
    - Star moves -> shifting light -> Doppler effect
      - Closer -> light waves scrunch! -> Blueshift
      - Away -> light waves stretch! -> Redshift
  - Transits
    - Focus

- Direct Imaging
  - Taking pictures
  - Problem: blocking out light from their host stars
- Gravitational Microlensing
  - Einstein…and how big objects warp the fabric of space!
  - Light distorts from intense gravity
  - Lensing events -> “blips” of light! -> light from a distant star is magnified by a planet’s gravity and it temporarily appears brighter than it typically does
  - Rogue exoplanets
- Astrometry
  - …Sort of like radial velocity?
  - Change in position (relative to other stars) rather than light
  - Star “wriggles”, look for exoplanet movement/wobbles

★ **Slide 4: An Exoplanet Transits When…**

- “Eclipse”/”Eclipse-ing” event
- Does not work on exoplanets that orbit at an angle not visible/accessible from Earth
- Atmospheric composition
  - Light shines through the planetary atmosphere
  - Atmospheric molecules absorb starlight
  - Identify chemical composition from the filtered light (spectroscopy)
    - Water and oxygen
- Using mass of the star & period to calculate the orbit
  - Kepler’s Laws are essential
- “By comparing the fraction of light that disappears during a transit to the total amount of light that typically emanates from a star, researchers can figure out the precise size of a planet. For example, if the starlight dims by 1 percent, this indicates that the planet is 1 percent the size of its star. Researchers estimate a star’s size by studying the overall spectrum of light produced by the star.” (MIT)

★ **Slide 5: What year was the first discovery of an exoplanet through the transit method?**

- Active participation 1
- Answer: B, 2002

★ **Slide 6: A Brief History**

- Follow-up to the active participation

★ **Slide 7: Transit Method Sparknotes**

- Mostly from a Ground-based telescope perspective
- Astronomical observations/imaging of a star over the duration of several hours
- Calibration -> “clean images” -> remove dust rings, CCD/camera idiosyncrasies or “deformities”, and sort of stationary disturbance
  - Clouds/weather interferences cannot be removed via this method
- Photometry -> transit curve -> change in brightness from passing planet over time

- ★ **Slide 8: Transit Curve**
    - Introduce each component one at a time
- ★ **Slide 9: Try It Yourself!**
    - Active participation 2
    - Worksheet
        - Puzzle or sorting transits to their respective exoplanets
        - DIY exoplanet
    - Work in groups
    - ~2-3 people get up and talk about their DIY exoplanet
        - Moonpie reward!
- ★ **Slide 10: To Conclude…**
    - Closure
- ★ **Slide 11: Questions**
- ★ **Slide 12: Links**

Self-Evaluation

Overall, I believe that the lesson went well! Some of the students from the class approached me after the lesson and gave me good feedback. It was reassuring because I was worried about the lesson's reception in the moment of presenting. However, it seemed like the students understood the lesson, which means the objective was fulfilled– this is not too much of an assumption I think, since it was semi-confirmed by intel from students. As for me, I get extremely anxious when I present in front of people, so I'm glad it did not interfere too much with the quality of the lesson and the students' understanding of it.

- **Pros**
    - Using questions in the PowerPoint helped me gauge the students' understanding in increments. There were moments when there was hesitance/answers were a mixed bag, but it is probably due to asking a question before presenting information on it. This may have caused students to feel uncertain about their answers. However, most of the engagement required information already covered by Dr. Rodruck in his lesson, so I was hoping students would reference their notes. Fortunately, I observed some of them flipping through their notes when confronted with a question.
    - The presentation was a little campy, which I personally enjoy; however, it was not entirely professional and could distract students. None of the students that I talked with said anything negative about the slides but, the overarching comment was that it was "engaging" and "interesting". I am unsure if those feelings resonate with the content, the presentation, or both, so I hope/like to believe the colorfulness was a helpful component of the lesson.
- **Negative**
    - I expected the moonpie idea to be better received and unfortunately, it was not. It was a little bit of a bummer and more notably, straight-up jarring to see college students not fiend at the opportunity for free food.
    - The students seemed to feel awkward and I am the most awkward person in the world. The vibes were awkward, the atmosphere was awkward. It might have been a battle to the death of awkwardness. Presenting feels awkward in general, so I could be biased. I do not know how to fix this.